

Multiple Realizability and Universality

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ABSTRACT

This paper concerns what Jerry Fodor calls a ‘metaphysical mystery’: How can there be macroregularities that are realized by wildly heterogeneous lower level mechanisms? But the answer to this question is not as mysterious as many, including Jaegwon Kim, Ned Block, and Jerry Fodor might think. The multiple realizability of the properties of the special sciences such as psychology is best understood as a kind of universality, where ‘universality’ is used in the technical sense one finds in the physics literature. It is argued that the same explanatory strategy used by physicists to provide understanding of universal behavior in physics can be used to explain how special science properties can be heterogeneously multiply realized.

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1 Introduction

In ‘Multiple Realization and the Metaphysics of Reduction’, Jaegwon Kim ([1992]) presents his, by now well-known, criticism of nonreductive physicalist approaches to the philosophy of mind. His target here is limited to those arguments against reductionism that depend essentially on the ‘Multiple Realization Thesis’—that upper level psychological properties or states are potentially realized by (or implemented in) a wide variety of *heterogeneous* lower level physical states.¹ Kim argues that the Multiple Realization Thesis

¹ For the purposes of the argument here, I’m just going to assume that the realization (or supervenience) base consists of physical properties, though of course it may be better to focus on neurophysiological realizers of the mental.

does *not* necessarily entail the failure of psycho-physical reductionism.² I do not wish, fully, to enter the reductionism debate here. Instead, I'd like to ask some questions about the nature of multiple realizability. In particular, I will try to offer a physically acceptable strategy for explaining how at least some upper level properties might be multiply realized by lower level physical properties.

Several authors, notably Jerry Fodor and Ned Block, have offered diagnoses of the intuition motivating Kim's argument. Fodor ([1997], p. 159), for instance, says that 'what's really bugging Kim is [...] a metaphysical mystery about functionalism.' It's best to quote Fodor here at some length.

The very *existence* of the special sciences testifies to reliable macrolevel regularities that are realized by mechanisms whose physical substance is quite typically heterogeneous [...] Damn near everything we know about the world suggests that unimaginably complicated to-ings and fro-ings of bits and pieces at the extreme *microlevel* manage somehow to converge on stable *macrolevel* properties.

On the other hand, the 'somehow' really is entirely mysterious, and my guess is that that is what is bugging Kim [...] [Kim] doesn't see why there should be (how there could be) [macrolevel regularities] unless, at a minimum, macrolevel kinds are *homogeneous* in respect of their microlevel constitution. Which, however, functionalists in psychology, biology, geology, and elsewhere, keep claiming that they typically aren't (Fodor [1997], pp. 160–1).

Fodor, as well, confesses to ignorance about the 'somehow' here. In fact, he says that he doesn't even 'know how to *think about why*' there could be such macrolevel regularities multiply realized by heterogeneous microlevel properties—why, as he puts it, 'there is anything except physics' (Fodor [1997], p. 161).

The mystery then is really how multiple realizability is possible. I hope to show that the situation is not really as mysterious from a physicalist perspective as Fodor (and presumably Kim) takes it to be. There is, in fact, a way to think about how such macroregularities are possible despite (maybe even in virtue of!) the complicated to-ings and fro-ings at the microlevel. In order to accomplish this I need to discuss a kind of explanation, explicit reference to which is becoming prevalent in the physics literature, but which I believe has received too little attention from philosophers of science. I'm referring here to the explanation of what is called universal behavior or universality. (I will explain shortly the special use of these terms by physicists.) I will need to argue (i) that the phenomenon of multiple realizability is

² Kim makes a distinction between those mental predicates that are functionalizable and, by his lights, genuinely pick out properties, and those which may not be functionalizable such as qualia. For these latter, while they may be in some sense multiply realizable, the issue of reduction will not arise.

itself an instance of universality, and (ii) that it is reasonable to expect that when very broadly construed the *same strategy* (though likely not the very same explanatory apparatus) can be used in dispelling the mystery to which Fodor refers.

In its broadest construal the explanatory strategy is ubiquitous; perhaps almost too ‘obvious’ to have been noted explicitly. However, physicists and mathematicians have done much to make precise a specific version of this general explanatory strategy. It is called the renormalization group (RG). It is worth exploring some of the specific details of the RG analysis to get a clear idea of the general features of this type of explanation.

Before getting to this, let me say a few things about Kim’s argument.

2 Jade, pain, microstructure and causal powers

Kim discusses a case of purported multiple realizability in which jade has two distinct physical realizers: jadeite and nephrite.³ He argues that since jadeite and nephrite are distinct chemical kinds we shouldn’t think of jade itself as a mineral natural kind.

But why is their being distinct as chemical kinds relevant here? Because many important properties of minerals, we think, are supervenient on and explainable in terms of, their microstructure, and chemical kinds constitute a microstructural taxonomy that is explanatorily rich and powerful. Microstructure is important, in short, because macrophysical properties of substances are determined by microstructure (Kim [1992], p. 14).

If pain is also realized by heterogeneous collection of physical properties, it too, according to Kim, will fail to be a genuine kind. Kim imagines that pain has three physically distinct realizations: N_h , N_r , and N_m —the realizations of pain in humans, reptiles, and martians. Suppose we define a property N by disjoining these three heterogeneous realizers. Kim poses the following problem.

If pain is nomically equivalent to N , the property claimed to be wildly disjunctive and obviously nonnominal, *why isn’t pain itself equally heterogeneous and nonnominal as a kind?* Why isn’t pain’s relationship to its realization bases, N_h , N_r , and N_m analogous to jade’s relationship to jadeite and nephrite? If jade turns out to be nonnominal on account of its dual ‘realizations’ in distinct microstructures, why doesn’t the same fate befall pain? (Kim [1992], p. 15).

As is well known, Kim’s argument for this conclusion depends in part on the failure of the projectibility of generalizations about pain: since all of the evidence for such a generalization may have come from investigation of

³ This example was first introduced by H. Putnam ([1980], p. 241) in ‘The Meaning of “Meaning.”’

humans, this would give us no reason to expect that the generalization would hold for reptiles or martians.

Now, Kim takes this argument to be indirect and analogical. He also offers a ‘direct argument’ that appeals to the ‘Principle of Causal Individuation of Kinds’:

Kinds in science are individuated on the basis of causal powers; that is, objects and events fall under a kind, or share a property insofar as they have similar causal powers (Kim [1992], p. 17).

This principle, in conjunction with what he calls the ‘Causal Inheritance Principle’ entails that multiply realized mental kinds cannot be ‘proper scientific kinds.’ The Causal Inheritance Principle states that if a mental property is realized on an occasion by a given ‘physical realization base *P*’, then the causal powers of the mental property, on this occasion, just are the causal powers of the particular realization base *P* (Kim [1992], p. 18).

Now, I take it that Kim must be thinking of causal powers as sets of atomic or molecular properties (microstructures) governed by certain specific force laws, etc. After all, he claims that jadeite and nephrite are distinct chemical kinds in virtue of their distinct microstructures. Recall, he says that ‘[m]icrostructure is important [...] because macrophysical properties of substances are determined by microstructure’ (Kim [1992], p. 14). Of course, this isn’t to say that the microstructure *causes* the macroproperties. Nevertheless, the causal powers of the macroproperties (if there are any such powers) are to be identified with the microphysical structure of the realizer upon a given occasion.⁴

So the idea is that instances of a mental property *M*

that are realized by the same physical base must be grouped under one kind, since *ex hypothesi* the physical base is a causal kind; and instances of *M* with different realization bases must be grouped under distinct kinds, since, again *ex hypothesi*, these realization bases are distinct as causal kinds. Given that mental kinds are realized by diverse physical causal kinds, therefore, it follows that mental kinds are not causal kinds, and hence are disqualified as proper scientific kinds (Kim [1992], p. 18).

I do not have a metaphysical theory about the nature of causal powers to offer. On the other hand, I do think that one way of seeing an answer to what Fodor calls ‘Kim’s worry’—how it is possible for macrolevel regularities to be *heterogeneously* multiply realized—will involve denying the identification of the ‘causal powers’ of the upper level property with the *detailed* microstructural makeup of its physical realizer upon any given occasion. To avoid misconceptions, I need to be careful here. I am definitely *not* claiming that the causal powers of the upper level property are not somehow to be

⁴ As far as I can tell, Kim’s argument won’t go through unless the causal properties of the macroproperties *just are* the resultant (or ‘sum’) of the microstructural properties.

grounded in the physical realization base. To do so would, I believe, be tantamount to denying what Kim calls ‘the causal closure of the physical domain.’ However, I am claiming that *most* of the details of the microstructure—those details that differentiate one realizer from another and allow us, legitimately, to speak of *heterogeneous* realizations—are (contrary to Kim) irrelevant to the causal powers of the upper level property.

Kim holds that the Principle of Causal Individuation of Kinds requires that we interpret the claim that e.g. ‘ N_h , N_r and N_m are heterogeneous as kinds’ to mean that ‘they are *heterogeneous as causal powers*’ (Kim [1992], p. 17). My claim is that this is not the right way to think about it. The properties N_h , N_r and N_m may very well be heterogeneous as kinds, but it doesn’t follow that ‘pain itself can show no more unity as a causal power than $N_h \vee N_r \vee N_m$ ’ (Kim [1992], p. 17). The ‘causal powers’ of pain (whatever they really are) are not necessarily grounded in something physically different each time pain is realized by a physically distinct set of microstructures. It is this possibility that I would like to show admits of a physically acceptable explanation.

Block ([1997], p. 120) argues that the realizers of the upper level properties, while heterogeneous, aren’t *completely* heterogeneous. ‘One way of leading into my disagreement with Kim is to note that these special science kinds are typically not nomically coextensive with completely *heterogeneous* disjunctions of physico-chemical properties.’ In support of this, he appeals to what he calls the ‘Disney Principle’ (Block [1997], p. 120). This is the claim that ‘laws of nature impose constraints on ways of making something that satisfies a certain description.’⁵ On his view it would ‘be amazing if laws of nature imposed *no constraints at all* on what can think or be conscious [or, presumably, on what can feel pain]’ (Block [1997], p. 120). The idea is that such constraints require some sort of similarity between the distinct realizers of the special science properties; and so, the realization base is not as heterogeneous as Kim and others might think.

On my view there is indeed something to the Disney Principle. But as an assertion of fact about the world it stands in need of explanation and justification. Without this explanation, it amounts to little more than a denial of Kim’s assertion that the physical realizers of the upper level property are so disparate in physical constitution that the disjunctive property nomically correlated with the upper level property fails to be a genuine kind. In other words, one would like to have some independent account of why the Disney principle holds. Block’s intuitions, I believe, are correct. But there ought to be a physically acceptable explanation of why. Simply claiming that it would be ‘amazing’ if things were otherwise, is not sufficient.

⁵ The principle is introduced as follows: ‘In Walt Disney movies, teacups think and talk, but in the real world, anything that can do those things needs more structure than a teacup’ (Block [1997], p. 120). Whence the name.

I will argue that it is reasonable to think of the macrolevel regularities as universal in the technical sense noted above, and that it is possible to say something positive about explaining that universality in a physically acceptable way despite the heterogeneity of the physical realizers. In effect, this should explicitly provide a justification of Block's use of the Disney Principle.

Let me now turn to a discussion of the nature of universality.

3 Universality

A broad goal of scientific theorizing is to recognize and explain observed patterns in the behavior of systems of a given type. This often, especially for systems of any appreciable 'size,' involves finding the dominant parameters that characterize the systems' behavior. One thereby delimits these dominant or important parameters from the multitude of other parameters required for a full characterization of the systems. The irrelevance of these other parameters is ultimately responsible for the reproducibility of experiments on individual systems as well as the ability to generalize experimental results to other, relevantly similar systems.

A natural extension of this aim is to see how much, and what types, of observed behavior can be generalized across distinct kinds of systems. Are there parameters that somehow characterize the dominant features of systems of widely different types? If there are, physicists often call the observed behavior and the dominant features 'universal'.

Most broadly construed, universality concerns similarities in the behaviors of diverse systems. As Michael Berry ([1987], p. 185) puts it, saying a property is a universal feature of a system is 'the slightly pretentious way in which physicists denote identical behavior in different systems. The most familiar example from physics involves thermodynamics near critical points.' It is in thermodynamics and statistical mechanics that the term 'universality' receives the most use in physics.

To begin to get an idea about the concept of universality as well as its ubiquity, consider the following homely example. One wants to explain the observed common behavior of pendulums—one wants, for example, to understand why pendulums with bobs of different colors, rods of different lengths, different masses composed of diverse materials, etc., all have periods (for small oscillations) that are directly proportional to the square root of the length of the rod from which the bob is hanging. At one level the explanation is quite straightforward: one solves the very simple equation of motion for such a system. The argument can be found near the beginning of almost any elementary textbook in classical mechanics. This is an entirely satisfactory account. We *know* that the relevant features are the acceleration due to gravity acting on the bob and the length of the rod; simple mathematics gives us the rest. But

there is another why-question which is far from simple. Why, one might ask, are factors such as the color and (to a large degree) the constitution or microstructural makeup of the bobs irrelevant for answering our why-question about the period of the pendulums? Why is *this* equation, rather than one in which, say, a color parameter plays a prominent role, explanatory? In other words, what allows us to bracket, or set aside as ‘noise’ these other features of the individual pendulums as inessential or irrelevant for the explanation of the behavior of interest?

These latter questions concern the explanation of universal behavior. They may seem somewhat silly in the case of understanding the behavior of pendulums, because we already have a *theory*—classical mechanics—that tells us what variables or parameters are relevant for specifying the mechanical state of a pendulum. More interesting cases arise, as we will see, when many (if not most) of the variables our theory requires for a complete state description are not relevant for explaining the behavior of interest. In fact, specifying all of the details required for a complete state description may actually detract from, or even *block* the desired explanation. However, were one to think quasi-historically about the development of classical mechanics, one would see that there really is no difference between the example of the pendulum and some of the more interesting cases we will consider shortly. Prior to settling what counts as the state of a classical system, the questions about the relevance of color, say, to the behavior of pendulums would not have seemed so silly.

Certain limit theorems of probability theory provide a more precise *example* of universality. Such theorems describe certain asymptotic regularities of mass, or collective behavior that obtain, in effect, independently of the details of the contributions from the individuals in the collective. For example, let μ be the number of occurrences of an event (e.g. heads) in n independent trials for which the probability of the event on each trial is p where $0 < p < 1$. The De Moivre–Laplace limit theorem states that

$$\lim_{n \rightarrow \infty} P \left\{ a \leq \frac{\mu - np}{\sqrt{np(1-p)}} \leq b \right\} = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{z^2}{2}} dz. \quad (1)$$

In other words, it states that the scaled and normalized random variable μ has a Gaussian or normal distribution. This theorem describes the collective behavior of a sequence of independent trials, where the individual contributions to the limit become more and more insignificant as the number of trials increases. This is an important feature—the asymptotic regularity displayed by the normal distribution law is in essence independent of the details of the individual trials. An application of this theorem might be to a sequence of independent measurements of a particular quantity performed on a gas. There are many factors that actually determine the state of the gas at the time of each

measurement and so determine the actual value of the quantity at any time. But as long as these various factors are independent, and individually insignificant with respect to the measured value of the quantity (the random variable), the result of many such measurements will display the familiar normal distribution or bell curve.

So, limit theorems of the De Moivre–Laplace sort give expression to the fact that the dominant features are collective features and the individual values are, by and large, irrelevant.

But there is a deeper, more important aspect to such limit theorems that is relevant to our discussion here. Suppose one wonders what the distribution of the appropriate random variable, μ , would be if the individual ‘factors’ were not independent of one another. In other words, what would happen to the *form* of the limit law if certain kinds of correlations are allowed between the outcomes of the individual trials. The simplest kind of correlation is represented by the Markov property, in which the probability of a particular outcome depends on the result of the previous trial. Mathematically this is represented by a transition matrix. One finds that the following limit law holds:

$$\lim_{n \rightarrow \infty} P \left\{ a \leq \frac{\mu - np}{\sqrt{np(1-p)C}} \leq b \right\} = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-\frac{z^2}{2}} dz, \quad (2)$$

where C depends on the transition matrix, n is the number of transitions, and p is the probability for the initial occurrence of a certain outcome. The important feature here is that the *same* asymptotic regularity is obtained even for the very different case in which we allow correlations, if the normalization of the random variable is appropriately altered. *In other words, the right hand sides of equations (1) and (2) are identical.* This is an instance of universality. The alteration (in this case the appearance of C) depends upon the correlations.⁶ Sinai notes that ‘[t]he appearance and the universality of the Gaussian distribution [...] can seem rather enigmatic and surprising’ (Sinai [1992], p. 130). As it turns out, the explanation for this universality is provided by (a probabilistic version of) the RG argument to be discussed below.

These limit theorems describe behavior that is largely independent of the details of the ‘interactions’ at the level of the individual components. As we shall see this feature is also present in critical phenomena. In fact, the connection between critical phenomena and limit theorems of probability is much deeper than a simple analogy. There are rigorous results connecting the two domains. (See Sinai [1982]).

As a final example consider the thermodynamics of various systems near their critical points. Experimentally it is found that certain dimensionless

⁶ The conceptual importance of this example is stressed in Jona-Lasinio ([1975]). See also my discussion in Batterman ([1998b]).

numbers, so-called ‘critical exponents,’ apparently characterize the (virtually) identical behavior of systems as diverse as fluids, and magnets near their respective critical points. The critical point for a fluid in a container, for instance, is a point below which it is impossible to pass, by increasing pressure at fixed temperature, from the gaseous phase to the liquid phase without going through a state in which both gas and liquid are simultaneously present in the container; and above which such a transition from gas to liquid is possible without passing through a regime in which both gas and liquid are simultaneously present. The critical exponent describes the behavior of the density of the fluid as a function of temperature near the critical point. It describes a universal property in that the same behavior is observed almost without regard for the chemical constitution of the fluid. More remarkable, still, is the fact that the *same* critical exponent apparently describes the behavior of magnets as they undergo a transition from the ferromagnetic state with positive net magnetization below the critical point, to the paramagnetic phase with zero magnetization above the critical point. Surely, one would like an explanation of this universal behavior. Modern statistical physics (RG arguments), provides such an account, more details of which will be discussed below and in the appendix.

From these examples it is possible to extract two general features characteristic of universal behavior or universality. First, the *details* of the microstructure of a given fluid are largely irrelevant for describing the behavior of the particular system of interest.⁷ Second, many different systems (other fluids and even magnets) with distinct microstructures exhibit identical behavior characterized by the same critical exponent.⁸ The first feature, arguably, is responsible for the second. Thus, as we will see, the RG analysis of critical phenomena will explain why the different systems exhibit the same behavior (the second feature) by utilizing the fact that at the critical point, the details of the microstructure of any one individual system are virtually irrelevant for *its* behavior at that point (the first feature).

My claim is that the same two features are also present in many of the examples of multiple realizability discussed in the philosophy of mind literature and in philosophical discussions of the status of the special sciences in general. It makes sense, I believe, to think of multiple realizability as an instance of universality. According to the Multiple Realization Thesis, generalizations about pain, for example, may be grounded on any given occasion in physically distinct microstructures. These generalizations are analogous to claims about the behavior of fluids at their critical points being

⁷ In the case of the limit theorems, the analog of the details of the microstructure is the details of the individual contributions to the collective behavior in a given sequence of trials.

⁸ Again, in the case of the limit theorems the analog here is the fact that the same, Gaussian, distribution is the limiting distribution.

characterized by the same critical exponent. They represent observed *patterns* of behavior present in distinct kinds of systems. The puzzle about multiple realizability (that is, the ‘metaphysical mystery about functionalism’ to which Fodor refers) is just the difficulty of explaining (or seeing how one might attempt to explain) a particular instance of universality—identical behavior displayed by distinct systems.

Part of what may make this seem so completely mysterious is the fact that philosophers by and large ignore the possibility that there are *physical reasons why the details of the makeup of the individual realizers may be largely irrelevant for the upper level behavior of the system*. That they ignore this possibility isn’t really surprising given the structure of the debate. On the one hand, discussions of functionalism typically eschew mention of the physical details of the realizers. Functionalism, after all, was developed in part with that as a desideratum. As a result, the consideration of such reasons—*physical* reasons for the irrelevancy of the physical details—simply doesn’t arise. On the other hand, partly in response to the functionalist point of view people have been led to emphasize the relevance of physical details. There is the often implicit, though sometimes explicit assumption/assertion that the microstructural details *are* crucial. This is prevalent in reductionist critiques of the nonreductive physicalist position. I have tried to show above how this assumption/assertion is crucial to Kim’s argument that causal powers and microstructure go hand in hand.

I suggest that this is the wrong way to think about it. The debate is focused on two extremes. The proper response may be to take a more moderate position. On my view, the physical makeup of the base properties are indeed relevant, contrary to the motivations of functionalism. But most of the microstructural details that rightly warrant the claim that the realizers are ‘wildly heterogeneous,’ are by and large irrelevant for the behavior of interest. In what follows I will try to show how this seemingly paradoxical position can be maintained. Furthermore, I will argue that there is an explanatory strategy that will allow us to think about why (to use Fodor’s colorful phrase) ‘there is anything but physics.’

4 Explaining universality

In this section I will discuss a particular example of the general strategy for explaining universality. This example concerns the physicist’s explanation of the universality of so-called ‘critical phenomena.’ As I noted earlier this is a fairly precise mathematical implementation of what I take to be a broadly applicable strategy for explaining universal behavior. My goal here is to outline briefly the key features involved in the explanation. A more detailed, though still qualitative discussion is included in the appendix.

To see how the explanation is supposed to go, we must first introduce a few more details about critical phenomena in systems such as fluids, gases, and magnets.

The critical behavior of a fluid is best characterized (at least at the simplest level) in terms of what is called the ‘order parameter.’ In this case the order parameter is the difference in densities of the vapor and liquid present in the container of ‘fluid’: $\Psi = \rho_{liq} - \rho_{vap}$. Below the critical temperature T_c , Ψ is nonzero indicating the simultaneous presence of the two phases of the fluid; while above T_c the order parameter Ψ vanishes. Figure 1 illustrates the key points. In particular, the vertical lines tie together values for coexisting liquid and vapor densities, (ρ_{liq}) and (ρ_{vap}) respectively.⁹

One universal aspect of the behavior of fluids near the critical point is the shape of the coexistence curve drawn in Figure 1. The key question concerns the shape of this curve near the critical temperature T_c . To begin to answer this one considers the ‘reduced’ temperature

$$t = \frac{T - T_c}{T_c}$$

which gives the difference in temperature from the critical temperature in dimensionless units. One makes the assumption (well supported by experimental data) that as T approaches T_c from below, the order parameter Ψ vanishes as some power β of $|t|$:

$$\Psi = \rho_{liq} - \rho_{vap} \sim |t|^\beta.$$

If the coexistence curve were parabolic near T_c , then β would be equal to 1/2. Experiments have shown, however, that the exponent β is not equal to 1/2, but rather has a value near 0.33 that is also, apparently, not a simple rational fraction. Furthermore, the *same* number appears to characterize the coexistence curves for different fluids and even, as noted earlier, coexistence curves for magnets where the order parameter is the net magnetization, M . The magnetization is positive below the critical temperature T_c , the system being in a ferromagnetic phase and is zero above T_c with the system in a paramagnetic phase:

$$\Psi = M \sim |t|^\beta.$$

One might ask about a given fluid, say F , why near the critical temperature its density varies with temperature the way that we observe it to. We might attempt to answer this starting from fundamental physics by appealing to the Schrödinger equation for the fluid considered to be some sort of collection of molecules interacting according to certain specified force laws, etc. If we were lucky in our attempt to solve the equation we might, in fact, be able to derive

⁹ Note that the vertical ‘tie lines’ in Figure 1 are *not* part of the ‘coexistence curve.’

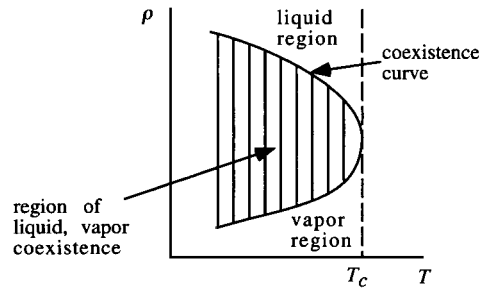


Fig 1. The coexistence curve: density vs. temperature for a fluid.

the fact that for fluid F , $\Psi_F \sim |t|^{0.33}$. On the other hand, such a derivation is *not* an explanation of the universality we are discussing. In no way can it answer the question of why the same exponent describes the critical behavior of fluids F' , F'' , etc.

The desired explanation is provided by the RG. To see how this goes, first note that every system, each different fluid say, is represented by a function—its so-called Hamiltonian. This characterizes the kinds of interactions between the system's components (or degrees of freedom), the effects of external fields, etc. Usually, the different components of a gas or fluid are correlated with one another only very weakly. In other words, they interact significantly only with nearby components and they remain virtually uncorrelated with remote components. However, when a system is near its critical point the 'length' of the range of correlations between the different components increases despite the fact that the interactions remain local. In fact, at the critical point, this so-called correlation length diverges to infinity. The divergence of the correlation length is intimately associated with the system's critical behavior. It means that correlations at every length scale (between near as well as extremely distant components) contribute to the physics of the system as it undergoes a phase transition. This is a highly singular mathematical problem and is, in effect, completely intractable. It is relatively easy to deal with correlations that obtain between *pairs* of particles, but when one must consider correlations between three or more—even more than 10^{23} —particules, the situation is hopeless. The guiding idea of the renormalization group method is to find a way of turning this singular problem into a tractable, regular problem by transforming the problem into an analysis of the topological structure of an appropriate abstract space—the space of Hamiltonians.

One introduces a transformation on this space that changes an initial physical Hamiltonian describing a real system into another Hamiltonian in the space. The transformation preserves, to some extent, the form of the original Hamiltonian so that when the thermodynamic parameters are properly

adjusted (renormalized) the new renormalized Hamiltonian describes a system exhibiting similar behavior. Most importantly, however, the transformation effects a reduction in the number of coupled components or degrees of freedom within the correlation length. Thus, the new renormalized Hamiltonian describes a system that presents a more tractable problem and is easier to deal with. By repeated application of this renormalization group transformation the problem becomes more and more tractable until one hopefully can solve the problem by relatively simple methods. *In effect the renormalization group transformation eliminates degrees of freedom (microscopic details) that are inessential or irrelevant for characterizing the system's behavior at criticality.*

In fact, if the initial Hamiltonian describes a system at criticality, then each renormalized Hamiltonian must also be at criticality (see the appendix). The sequence of Hamiltonians thus generated defines a trajectory in the abstract space that in the limit as the number of transformations goes to infinity ends at a fixed point. The behavior of trajectories in the neighborhood of the fixed point can be determined by an analysis of the stability properties of the fixed point. *This analysis also allows for the calculation of the critical exponents characterizing the critical behavior of the system.* It turns out that *different physical* Hamiltonians can flow to the same fixed point. Thus, their critical behaviors are characterized by the same critical exponents. This is the essence of the explanation for the universality of critical behavior. Hamiltonians describing different physical systems fall into the basin of attraction of the same renormalization group fixed point. This means that if one were to alter, even quite considerably, some of the basic features of a system (say from those of fluid F to fluid F'), the resulting system (F') will exhibit the same critical behavior. This *stability* under perturbation demonstrates that certain facts about the microconstituents of the systems are individually largely irrelevant for the systems' behaviors at criticality. Rather, their *collective properties* dominate their critical behavior, and these collective properties are characterized by the fixed points of the renormalization group transformation (and the local properties of the flow in the neighborhood of those points). We can, through this analysis, understand both how universality arises and why the diverse systems display identical behavior.

By telling us what (and why) various details are irrelevant for the behavior of interest, this same analysis also identifies those physical properties that *are* relevant for the universal behavior being investigated. For instance, it turns out that the critical exponent can be shown to depend on the spatial dimension of the system and on the symmetry properties of the order parameter. So, for example, systems with one spatial dimension or quasi-one dimensional systems such as polymers, exhibit different exponents than (quasi-) two dimensional systems like films. The universal behaviors of these systems

are, in turn, different than that exhibited by three dimensional systems such as fluids. The details here aren't crucial for understanding the main point: The RG type analysis illuminates those physical features that *are* relevant for the upper level universal behavior, *and at the same time demonstrates that all of the other details which distinguish the systems from one another are irrelevant.*

I don't think it is appropriate to speak of the spatial dimension of the system and the symmetry properties of the order parameter—the base properties that characterize the universality class—as *causal powers*. (Lacking any meta-physical theory about causation, I'm generally worried about causal power talk.) On the other hand, it does make sense, given this analysis, to hold that generalizations involving the upper level universal feature—the critical exponent β —*depend* essentially on these shared lower level features of the realizers. These features are, in effect, *nontrivial necessary conditions* for the upper level universal behavior of systems which are representable at the lower level by generalized Ising models with finite range interactions among their basic components.

5 The general explanatory strategy

What are the features of the RG explanation outlined in the last section that are likely to be characteristic of the general explanatory strategy? In the case just discussed, the explanation proceeds by developing reasons, grounded in the fundamental physics of fluids, for ignoring detailed microphysical information about any actual system or systems being investigated. It is a method for extracting structures that are, with respect to the behavior of interest, detail independent. Of course, certain features remain system or detail dependent. In the above example, each fluid will have a different critical temperature T_c that depends essentially on its microstructural constitution. Nevertheless, the RG method allows for a construction of a kind of limiting macroscopic phenomenology—an upper level generalization, if you will—that is not tied to any unique microstructural account. The microscopically distinct systems can be grouped into distinct universality classes depending on the macroscopic phenomenology they exhibit.

Nigel Goldenfeld *et al.* ([1989], p. 368) argue that the principles of RG theory constitute a 'general framework for extracting phenomenological relations among macroscopic observables from microscopic models that may not be precisely definable.' It is a

method to study the asymptotic behavior of a system, e.g., in the macroscopic limit, where the scale of observation is much larger than the scale of microscopic description. [...] It should be stressed that phenomenology is not a second-rate incomplete description of a given system, but the most complete description modulo microscopic unknown factors over which

we can never have ultimate detailed knowledge (Goldenfeld *et al.* [1989], p. 368).

While this last remark seems to reflect limitations on knowability, the RG type arguments are *not* essentially epistemic in nature. They demonstrate, as I've said, that the explanation of universal behavior involves the elucidation of principled physical reasons for bracketing (or setting aside as 'explanatory noise') many of the microscopic details that genuinely distinguish one system from another. In other words, it is a method for extracting just those features of systems, viewed macroscopically, which are stable under perturbation of their microscopic details.

As the example from the last section shows, it is essential that the microscopic details do not themselves greatly influence the phenomenological behavior of the system. For instance, the exact structure of the molecules in a given fluid and the precise nature of the forces characterizing the molecular interactions are not important for the critical behavior of that particular fluid. Instead, in the *asymptotic regime* where many molecules find themselves correlated—that is, where the correlation length diverges—it is the collective or correlated behavior that is salient and dominates the observed phenomenological behavior. In order to 'see' the stable, universal phenomenological structures, one must investigate the appropriate limiting regime. It is, therefore, an essential feature of the method that it describes the relationship between two asymptotically related descriptions of the system.¹⁰

In sum, the following features are apparently characteristic of RG type explanations of universal behavior.

- As the universal behavior is in some sense emergent in an asymptotic regime, the explanation involves some kind of asymptotic analysis.
- The universality is the result of the stability under perturbation of the underlying, subvenient, details. This is to say that the complex 'to-ings and fro-ings' at the microlevel are by and large irrelevant for the explanation.
- The stability under perturbation (or 'structural stability') is explained by the underlying microscopic physics of the systems being investigated.

That microstructurally different systems fall in the same universality or equivalence class, is the physicists' way of saying that the upper level universal behavior is multiply realized. And so, the explanation of the universality of the behavior is an explanation of the behavior's multiple realizability.

¹⁰ In many instances, the asymptotic relation between the different 'levels of description' will depend on differing length scales. But there is no essential reason that length (or size) must be the asymptotic parameter. Long time behavior or increasing complexity might, for example, be appropriate parameters for some investigations.

6 Special sciences

It remains to show that an explanatory strategy importantly similar to that just described, is applicable in those cases of interest to philosophers concerned with the multiple realizability of psychology in particular, and of the special sciences in general. This, of course, is the heart of the matter. Why should something like this explanatory strategy be at all likely to apply to higher level properties like pain, etc., which are not ordinarily taken to be properties of physics? The nice thing about the explanation of the universality of critical phenomena is that it takes place entirely within ‘the domain of physics.’ But, after all, psychology isn’t physics. Why should one think that there are facts about structural stability that explain why radically materially distinct systems can feel pain (assuming it’s a fact that they do)? Even worse, how can considerations of structural stability play a role in explaining how an ‘and’-gate can be instantiated in silicon, in hydraulics, and in cats, mice, and cheese (Block [1997], p. 120)?

The last question, of course, makes my proposal sound fairly inapposite. Perhaps, in the case of ‘and’-gates and other highly abstract properties, this sort of explanation for their multiple realizability will be neither suitable nor available. (Though, my own view is that the Disney principle needs justification even here.) But in these cases, the demand for an explanation doesn’t seem at all pressing. This may simply be because we already have an implicit appreciation of what is relevant and irrelevant for the universal behavior of interest. After all we have become quite adept at building systems which realize these abstract properties. On the other hand, as we’ve seen, Kim, Block and Fodor all seem to believe that there is an important why-question underlying many instances of multiple realizability in the special sciences. Recall Fodor’s remark about the mystery of how there could be macrolevel regularities which are not homogeneous with respect to their microlevel constitution. My only concern is to try to dispel some of this mystery. (Thus, my aim is not to provide the explanation for the multiple realizability of pain; instead, I’m only trying to show that such an explanation need not be physically mysterious.)

So the question remains as to how and whether facts about some kind of structural stability might be invoked to explain how wildly heterogeneous physical systems can feel pain. Block’s Disney principle is, as we have seen, a claim to the effect that such pain feeling systems cannot be completely radically distinct. If demonstrations of the sort we’ve been considering are relevant, then if one can provide such a demonstration, one has an explanation of the nature and extent of the constraints imposed on realization by ‘laws of nature’ (the laws of physics). This follows because the delimitation of universality classes is, in essence, a specification of just those constraints.

But again, what about the view that psychology (or any of the other special sciences) is not physics? Why should Block's Disney principle, even if it receives the desired physically grounded explanation, be applicable? To answer this, let me discuss briefly a paper by Mark Wilson ([1985]) which has, I believe, received too little attention in this literature. Wilson ([1985], p. 227) defends 'the old-fashioned "type/type" identity theory between mental and physical properties, i.e., the doctrine that mental attributes such as *is in pain* properly count as straightforward physical traits of the *has a mass of one gram* stripe.' Wilson is concerned to show that the 'neofunctionalist' arguments of e.g. Fodor ([1974]) in his original 'Special Sciences' paper rest largely on a confusion about the nature of physics' fundamental properties or 'kinds.' I don't intend to rehearse his arguments here. Instead, I want to discuss one major feature of his treatment that is relevant to the question raised immediately above—that is, why think that a physical explanation will be appropriate or even possible?

Wilson notes that physics *must* (to be able to do the work that it does) be very liberal about what counts as a physical property—so liberal, in fact, that just about any real-valued function of the state variables of a system must be a legitimate physical property of the system.

Suppose we represent the basic laws of (classical) physics in their Lagrangian form. Each system's behavior will be determined by a function, its so-called Lagrangian, which is essentially the difference between the kinetic energy T and the potential energy V of the systems:

$$L(\mathbf{q}, \dot{\mathbf{q}}, t) = T - V.$$

Here ' $\mathbf{q}$ ' and ' $\dot{\mathbf{q}}$ ' represent, respectively the cartesian position coordinates and velocity coordinates of the system's components, and ' t ' represents time. These are the system's state variables. Lagrange's equations of motion tell us how $L(\mathbf{q}, \dot{\mathbf{q}}, t)$ —and, hence, the system—evolves over time. Wilson notes that for all of this to work, physics requires an existence postulate ($\exists$) for physical properties having something like the following form (Wilson [1985], p. 232):

For any system S with Lagrangian L and any real-valued mathematical relation r , there is a physical trait ϕ such that $\phi(S, t)$ if and only if $r(\mathbf{q}(S, t), \dot{\mathbf{q}}(S, t), t)$.

Thus, for $\phi(S, t)$ to be a *physical property* of S it need only be the case that ϕ counts as a 'generalized' coordinate for S ; that is, that it satisfies Lagrange's equations:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) - \frac{\partial L}{\partial \phi} = 0.$$

As Wilson stresses, the important point is that ($\exists$) virtually guarantees (because

of its universal quantification over real-valued functions) that if system S feels pain, there exists some physical property ϕ_p nomologically coextensive with pain.

I've skipped many of the niceties of Wilson's discussion here, but the main point I want to highlight is there is no justification for dismissing the possibility of finding a physically based explanation for the multiple realizability of mental properties such as pain *merely* by appeal to the fact that psychology, after all, isn't physics. There may very well be grounds for doubting that the physical characterization of the property ϕ_p , guaranteed by $(\exists)$ to be coextensive with pain, is useful in any way for psychology. And, there may be grounds for doubting that the physical characterization of this property will play a crucial role in addressing the metaphysical worry about functionalism that is our primary concern. In fact, I think there are good reasons for skepticism on both of these scores. The demonstration that ϕ_p exists and isn't 'disjunctive' in the way Fodor, Kim, and others often insist is important, as Wilson shows, for clarifying various misunderstandings about the nature of physics' 'fundamental' properties. And although I doubt that ϕ_p will play much, if any, role in the explanation we seek, its existence indicates that we aren't completely off base in searching for some kind of physically grounded explanation of multiple realizability. In fact, given Wilson's argument our question about why the physically heterogeneous microstructural properties N_h , N_r , and N_m all realize pain is equivalent to asking why each of N_h , N_r , and N_m are realizers of ϕ_p ; and this question is entirely within the domain of the base (= physical) theory. This is true even if we might never be able to express ϕ_p in any reasonable language of physics.¹¹ Our interest is still in explaining the multiple realizability of pain.

It is important to emphasize that despite the fact that $(\exists)$ guarantees the existence of a physical property ϕ_p coextensive with pain, it is still the case, by hypothesis, that the realizations N_h , N_r , and N_m are heterogeneous. They are heterogeneous in the sense that their microstructures—their atomic and molecular constitution and their governing force laws—are quite different. And, it is with good reason that we distinguish the realizers from one another in terms of the distinct nature of their microstructures. These differences are often recognized by physics as being important. They are the same reasons we think of fluid F as fundamentally different from fluid F' . For example, it is because of these microstructural differences that F has critical temperature $T_c \neq T'_c$, the critical temperature of F' .

Another potential analogy between the two cases concerns the asymptotic nature of the emergence of the upper level properties. Here things are

¹¹ See Wilson ([1985], p. 232) for the details.

necessarily a bit speculative¹² and I'm not now going to defend this particular view of emergence. However, it does seem reasonable to suppose that the physical structures that are pain's various realizers are structures (states of the system) that are composed of many interacting parts. Connectionist models of computational features of the mind, for instance, require large numbers of interacting components. In fact, much of the mathematical apparatus of at least one class of network models is formally similar to the statistical mechanical modeling of lattice systems and spin glasses—just those models often used to represent the microscopic details of fluids, and magnets (see Churchland and Sejnowski [1992], Ch. 3). The mathematics of such models involves, essentially, the study of systems in the limit of a large number of interacting components.

I do not want to suggest that the mathematical apparatus of the RG analysis *itself* is what will provide us with an explanation of the multiple realizability of pain. I would have to know much more about the physics and microstructural makeup of the realizers. But, I do not see how any explanation would go unless it proceeds by demonstrating principled physical reasons for why most of the microstructural details of the realizers are irrelevant for the universal behavior of interest. The underlying physical theory must be able to tell us (most likely through some type of asymptotic analysis) that most (otherwise important) physical details are irrelevant.

Recall the brief discussion above about the universal behavior of pendulums. Over time physics has given us reason to believe that details like the color and, to a large extent, what the different bobs are made of—what their microscopic constitution is—are completely irrelevant for the manifest behavior of the pendulums. In so doing, physics has delimited those details that are relevant; namely, the length of the rod and the acceleration due to gravity. Physics has abstracted the structurally stable features of the observed universal behavior.

I think that we can imagine, if not expect, the same sort of thing in various cases of multiple realizability in the special sciences. Suppose that physics tells us that the physical parameters α and γ are the (only) relevant physical parameters for the pain universality class. That is, that N_h , N_r , and N_m have these features in common when certain generalizations or regularities about pain are the manifest behaviors of interest observed in each of humans, reptiles, and martians. Equivalently, physics has told us that all the other micro-details that legitimately let us think of N_h , N_r , and N_m as heterogeneous are irrelevant. We then have our explanation of pain's realizability by wildly diverse realizers.

We also have an explanation for the truth of Block's Disney principle. In

¹² Though, really they are no more speculative than the 'fact' that N_h , N_r , and N_m are all realizers of pain. This is a premise of the entire argument. Without it, there is no metaphysical worry to explain!

placing humans, reptiles, and martians in the same universality class characterized by α and γ physics will have demonstrated that there are constraints on what systems can realize pain.¹³

If we now want to switch to talk of causal powers, as Kim does, then it *might* be reasonable to associate the causal powers of the mental property pain with the physical features α and γ .¹⁴ In this way, despite the wild heterogeneity of the realizers N_h , N_r , and N_m , there will be no reason to think that pain's *causal powers* will likewise be wildly heterogeneous. In other words, the fact that pain's realizers— N_h , N_r , and N_m —are heterogeneous as physical kinds does not entail that pain is heterogeneous *vis-à-vis* its causal powers. Multiple realizability by diverse physical realizers does not entail, as Kim tries to argue, that the mental properties (and hence the special science itself) is also completely causally heterogeneous. Heterogeneity as kinds does not entail heterogeneity with respect to causal powers.

7 An objection and a reply

My position regarding multiple realizations in the special sciences and reduction is somewhat unusual. If one holds a fairly standard philosophical view of reduction (some refinement of Nagel's model, e.g.), one can maintain that the multiple realizability of some special science property results in a failure of reduction. This much is not unusual. But on many (perhaps most) such views of reduction, to reduce is to explain. My position is unusual because it asserts the possibility of providing physical explanations for why certain special science properties are multiply realized without also providing a reduction. One might object that I have made a fundamental mistake: multiple realizability just means a failure of reducibility and, hence, a failure of explanation. And, since I claim to be able to provide an explanation, all I have really done is show that reduction is possible and that the upper level property was not multiply realized after all.¹⁵

David Papineau ([1993], Ch. 2) appears to hold just such a view about the relations between explanation, reduction and multiple realizability. For Papineau, the reducibility of some special science property to physics requires 'only that there should be *some* physical property present in all and only [the distinct realizers of the special science property].' The presence of such a

¹³ As in the case of the fluids discussed above in Section 4, α and γ would be non-trivial, substantive necessary conditions for the multiply realized upper level property of pain. What, in this case, would be the analog to the characterization of the systems as those representable by generalized Ising models with finite range interactions, remains, of course, a matter for empirical investigation.

¹⁴ It might make more sense to speak of causal powers in this context (here it is a fairly common idiom), than it does in the context of critical phenomena discussed above.

¹⁵ Thanks to an anonymous referee for raising this worry in an acute form.

common physical property will then provide a ‘uniform physical explanation of why those instances always give rise to a certain sort of result’ (Papineau [1993], p. 35). For instance, it is the fact that physically diverse gases have a given mean kinetic energy that explains why they all have the same temperature. ‘It is this commonality that then enables us to *explain* such things as why an increase in temperature at constant volume always results in an increase in pressure’ (Papineau [1993], p. 35).

Papineau argues that the multiple realizability that (often) accompanies functionalism tells us that there can be no commonality which will yield the desired explanation. He finds this to be ‘incredible.’

The difficulty I am concerned with arises when some mental state *S*, which mediates between physical input *R* and physical output *T*, is realized by a range of different physical states *P_i*. The puzzle is: why do all the different *P_i*s which result from *R* all nevertheless yield the common effect *T*? (Papineau [1993], p. 35).

This is another way of putting Fodor’s metaphysical mystery about functionalism.

Papineau’s argument ([1993], p. 43) in Ch. 2 of *Philosophical Naturalism* suggests that ‘a finite disjunction of different physical states [the realizers] ought not to count as a reduction [of the associated realized macrolevel property] whenever the picture it leaves us with at the reducing level is physically *incredible*.’ In fact, he suggests ([1993], p. 43) that ‘the notion of a reduction is precisely the notion of an account which shows that nothing incredible is happening at the physical level.’ Since I claim to have shown how one might go about explaining such instances of multiple realizability, I have shown that nothing incredible is really going on at the physical level. But, on Papineau’s view, to have shown this is already to have provided a reduction, so my intermediate position would be untenable.

Suppose we grant the commonly held view that a necessary condition for reducing a macrolevel property is the identification of some lower level physical property that is both necessary and sufficient for that macrolevel property. The RG type analyses I have been discussing fail to satisfy this necessary condition for reduction. This is simply because those analyses do *not* actually provide such properties, nor were they intended to. Recall the explanation for the universality of critical phenomena in fluids discussed in Section 4. The explanation involves, essentially, the demonstration of stability under perturbation of the underlying subvenient physical details. In demonstrating this the RG analysis also illuminated certain physical features—the spatial dimension and the symmetry property of the order parameter—that are shared by systems in the universality class. These properties are *not sufficient* for a system to exhibit the upper level universal behavior characterized by the

critical exponent β ; hence, no reduction has been effected. Furthermore, the success of the explanation in no way requires that they be sufficient.

The explanation provided by the RG method in the case of universal critical behavior differs significantly in kind from Papineau's explanation of the temperature/pressure relation in a gas. For Papineau, one has an explanation *and hence, a reduction*, when one provides a common physical property—shared mean kinetic energy—that is *both* necessary and sufficient for the observed common upper level behavior. But, my point has been that the RG account explains the universality of critical phenomena without finding any such property. Explanations, then, can show that there is nothing physically mysterious or 'incredible' going on at the physical level without providing reductions.

8 Conclusion

I have been arguing that there is an explanatory strategy within which one might fruitfully address the question of how properties of the special sciences can be heterogeneously multiply realized. In physics, such explanations are common. In their most abstract form, they involve extracting phenomenological relations among upper level properties by determining features which are stable under perturbation of the micro-details. These arguments, as I've tried to show, are completely grounded in the lower level theory. They depend on a *demonstration* that from the point of view of the lower level theory many of the microstructurally relevant details are irrelevant for the upper level phenomenology of interest. In the physical cases there is a fairly well-understood set of methods—the renormalization group among them—which formalizes this kind of demonstration.

If, broadly construed, these considerations apply to the question of multiple realizability in psychology and in the philosophy of mind, then we can have a physically grounded explanation of psychology's multiple realizability without having to endorse a fully reductionist position such as one advocated by Kim. The idea here is that philosophical models of reduction, take reduction essentially to involve derivation along the lines initially proposed by Nagel. But we have seen reason to maintain the view that multiple realizability blocks this sort of reduction. There are good physical reasons for thinking of the different realizers as diverse as physical kinds. It is legitimate to treat the different systems as genuinely heterogeneous; and this does, for familiar reasons, make reduction difficult if not impossible.

On the other hand, it may be best, in this context, to give up on the various philosophical models of reduction which require the connection of kind predicates in the reduced theory with kind predicates in the reducing. Perhaps a more fruitful approach is to investigate asymptotic relations between the

different theory pairs. I do not have the space to develop these suggestions in detail here.¹⁶ My view is that in certain situations, the philosophical models of reduction that require formal derivations demand too much detail and, in some instances, too much precision. Sometimes, what is needed to understand relations between theories are methods for ignoring details. Such asymptotic methods often allow for the understanding of emergent structures which dominate observably repeatable behavior in the limiting domain between the theory pairs. I hope the above discussion in some way helps to make these statements worthy of further investigation.

9 Appendix: qualitative features of a renormalization group analysis

The following provides a more detailed, though still very qualitative, discussion of what is involved in the renormalization group explanation of the universality of critical behavior.

We have seen that the order parameter Ψ for a fluid is the difference in densities of the vapor and liquid in the container. Suppose we consider some region in the container and ask about the spatial extent of a ‘droplet’ of near liquid density in the surrounding sea of vapor. Because of fluctuations in density, this region may contain a liquid droplet of larger than average size. Nevertheless, for a fluid in equilibrium at a given temperature away from criticality, there is a well-defined average size for the droplets. One can now introduce the correlation length, ξ , which, roughly speaking, characterizes the spatial extent of the average droplets of liquid. (Put slightly differently: The correlation length is the typical distance over which the behavior of one microscopic variable or degree of freedom can be correlated with the behavior of another.) If all parameters other than temperature are fixed, ξ is a function of temperature, $\xi(T)$. A crucial fact is that at the critical temperature, the correlation length diverges to infinity:

$$\xi(T) \rightarrow \infty \text{ as } T \rightarrow T_c.$$

As another example consider a lattice of spins above the critical temperature T_c and in zero external magnetic field. Neighbouring spins on the lattice interact in a way that tends to align them parallel to each other. This accounts for the ferromagnetic nature of the lattice. At high temperatures, however, thermal energy tends to randomize the directions of the spins. Thus, while the ferromagnetic interaction tends to correlate the spins, at higher temperatures this tendency can be overridden, resulting in a loss of correlation. So, at high temperatures regions or blocks of spins will have a very small correlation

¹⁶ See my remarks concerning reduction in Batterman ([1995], [1993], [1997], [1998a]).

length, and the net magnetization will be zero. As the temperature decreases towards the critical or Curie temperature, the spatial extent of the blocks with correlated spins will increase. The correlation length, ξ , is a measure of the size of the correlated regions or blocks. As for the fluid, the correlation length is a function of T which diverges at the critical point—the point of transition from the paramagnetic phase (above T_c) to the ferromagnetic phase.

So near the critical temperature, the correlation length becomes extremely large. This means that an enormous number of degrees of freedom are coupled together. In other words, the range of the interaction among the degrees of freedom (e.g. the different spins in a correlated block of spins) is large. These correlations making solving the governing equations extremely difficult.

The idea behind the renormalization group methods is to relate an intractable problem involving a large correlation length to a more tractable problem characterized by a correlation length reduced by some factor. One is thereby reducing the number of coupled degrees of freedom. For example, suppose one has a system of spins on a lattice of spacing 1.0, as shown in Figure 2a) (round dots). Let us now group the spins into blocks of four spins. Next suppose that we have a way (via some kind of summation or partial integration) of replacing the grouped spins with a single new kind of spin—a ‘block’ spin—in the center of the block (square dots). The block spins, in this example, are now located on a lattice of spacing 2.0. Next, Figure 2b), one transforms the lengths so that the new spins are arranged on the same lattice sites as the original spins. In other words, all lengths are contracted by a factor of two. Finally, Figure 2c), one transforms the new spin variables so that the ‘new’ system of spins will be as much like the original system as possible.

As noted in section 4, each system is represented by a function—its so-called Hamiltonian. This function characterizes the kinds of interactions between the degrees of freedom (e.g. between the spins) as well as any effects of external fields, etc. Thus, the Hamiltonian can be considered to be a function of various system parameters—in particular, the temperature, the coupling constants characterizing the spin-spin interactions, and the external fields.¹⁷ So, Figure 2 in effect schematizes a transformation from one Hamiltonian to another in which, among other things, the correlation length of the new, transformed, Hamiltonian is smaller by the spatial rescaling factor which in this case is 2.

If one performs this ‘renormalization’ transformation (call it ‘ τ ’) repeatedly,

¹⁷ Strictly speaking, since this function is explicitly a function of the temperature, it is not truly a *Hamiltonian*. The idea is that temperature is just another parameter describing the system. It does not play any special role. So, whenever we speak of a system in this context, we mean a state of a system at a given temperature.

¹⁸ This process introduces all sorts of new couplings between the various degrees of freedom. Dealing with this increase of complexity is part of the ‘art’ of the method. It is not necessary to discuss, here, the various technicalities involved.

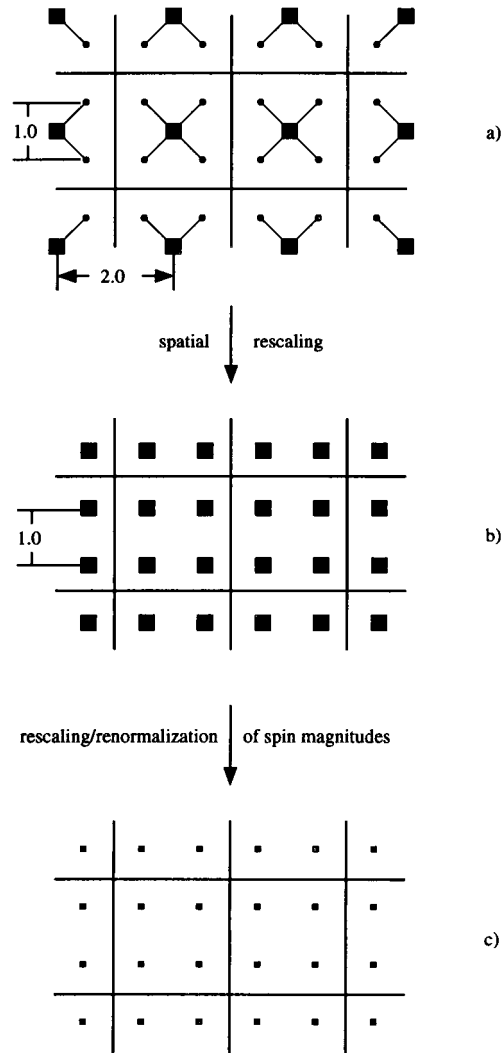


Fig 2. The operations of the renormalization group: a) block formation (partial integration); b) spatial rescaling/contraction; c) renormalization of parameters.

one gets a sequence of Hamiltonians all of which describe systems with the same lattice spacing, but where the correlation length gets smaller and smaller with each iteration. As a result, the new Hamiltonians describe systems with fewer and fewer coupled degrees of freedom within their correlation lengths. Thus, the systems are farther and farther from criticality.¹⁹

Now, let us consider an abstract space whose coordinates are the parameters

¹⁹ Note, this variation is *not* the renormalization group transformation, τ .

appearing in the various Hamiltonians of the systems. In effect, every point of this space corresponds to a possible Hamiltonian, so the space should be thought of as the space of all possible Hamiltonians. Consider, in particular, a lattice system undergoing a ferromagnetic phase transition. Suppose that the external fields and all of the coupling constants are fixed, and that the temperature is the only parameter being varied.²⁰ As the temperature approaches the critical temperature T_c , the point representing the system undergoing the phase transition moves about in the abstract space of Hamiltonians.²¹ Following the discussion in Pfeuty and Toulouse ([1977], pp. 12–5) let us call this path the ‘physical line’ of the system.

Each point on the physical line represents the system at a given temperature with a correlation length considered here to be solely a function of temperature: $\xi(T_i) = \xi_i$. One can now imagine this space of Hamiltonians divided up into surfaces of constant ξ . Figure 3 illustrates the general idea.²² The surface, S_∞ , is the ‘critical surface’ corresponding to parameter values (points in the space, i.e. Hamiltonians) having infinite correlation length. Under the renormalization group transformation τ every point on the physical line gets mapped to another point in the space of Hamiltonians. This yields a trajectory issuing from that point on the physical line. *An important feature here is that the trajectory generated by τ from the point p on the critical surface S_∞ remains confined to the critical surface.*

The reason for this is simply that under the renormalization transformation, lengths are contracted by a spatial rescaling factor ‘ b ’;²³ and

$$\tau(\xi_c) = \frac{\xi_c}{b} = \frac{\infty}{b} = \infty.$$

In other words, the correlation length remains infinite. The transformations of points other than p on the physical line yield trajectories that diverge from the critical surface S_∞ intersecting surfaces S_ξ corresponding to successively lower values of correlation length, thereby effecting the reduction of the number of degrees of freedom within a correlation length. This is illustrated in Figure 4.

The point p^* , shown in Figure 4 is called a fixed point. This is a point in the parameter space which, under τ , is its own trajectory. That is, it represents a state of a system which is invariant under the renormalization group transformation. Of necessity, such a fixed point has an infinite correlation length and so lies on the critical surface S_∞ .²⁴

²⁰ See footnote 17.

²¹ A very nice discussion of these issues, at a fairly intuitive level can be found in (Pfeuty and Toulouse [1997], Ch. 1).

²² In Figure 2, $b = 2$.

²³ Actually, a fixed point could have zero correlation length, since zero is also a length which is unchanged under division by b . Such fixed points are called ‘trivial,’ and are usually ignored.

²⁴ See Pfeuty and Toulouse ([1977], pp. 13–4).

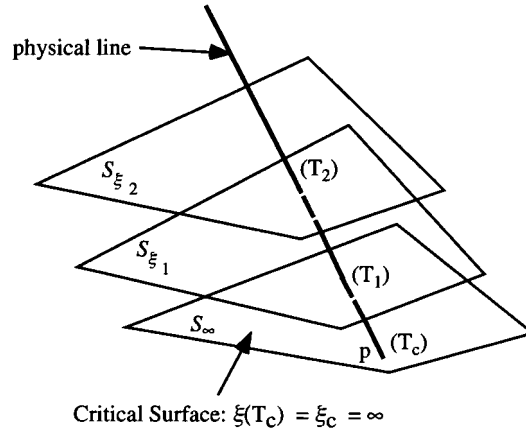


Fig 3. Surfaces of constant correlation length. $\xi(T_1) = \xi_1, \xi(T_2) = \xi_2$ where $\xi_1 > \xi_2$.

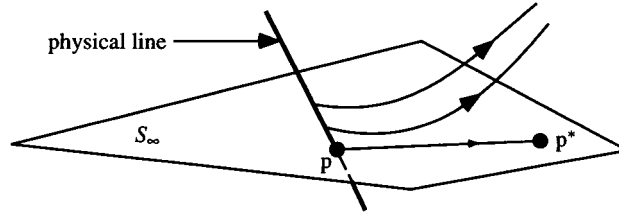


Fig 4. Renormalization group trajectories issuing from points on the physical line. p^* is a fixed point of the transformation τ .

We now have (almost!) sufficient detail to explain the universality of critical phenomena. Note that a fixed point of transformation is a property of the transformation itself. This means that to find them one must solve the fixed point equation:

$$\tau(\mathcal{H}^*) = \mathcal{H}^*.$$

That is, one must determine the fixed point Hamiltonian $\mathcal{H}^* = p^*$ and this is independent of any choice of initial Hamiltonian. Universal behavior is explained by reference to properties of certain fixed points. More precisely, it is related to the stability of the fixed points and to how, the renormalization group transformation τ maps points in the neighborhood of the fixed points. To illustrate, consider the simplest visualizable case.

Let us imagine that the critical surface S_{∞} is two dimensional. We want to consider the contour structure of the surface. Consider Figure 5.²⁵ One should think of the point b as if it were the top of a mountain. The trajectories of the

²⁵ See the discussion in Fisher ([1983], pp. 84–7).

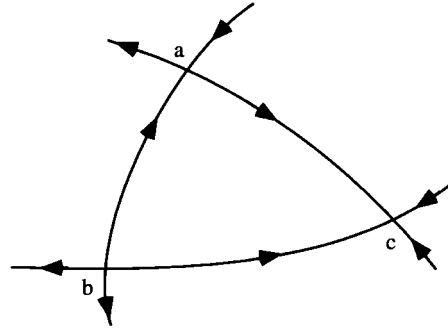


Fig 5. Possible fixed points on a two-dimensional critical surface.

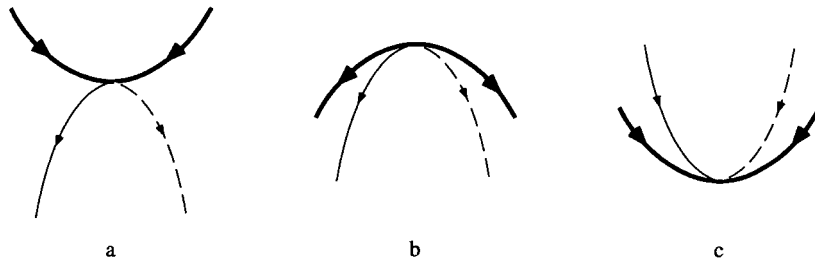


Fig 6. Contours near various fixed points on a two-dimensional critical surface.

renormalization group then are lines of steepest descent from the summit. Point c , on the other hand, is like the bottom of a valley. Finally, a represents a pass or saddle point.

Figure 6 provides another perspective of the same fixed points and the lines of steepest descent. It is clear that these fixed points are characterized by differing degrees of stability. For instance, the peak, b , is the least stable—any point in the parameter space near to, but not identical with b will flow, under τ , away from b . The valley, c , is the most stable—nearby points will be taken towards c by τ . The saddle, a , exhibits some sort of intermediate degree of stability, being stable in one direction and unstable in the other. These are the most important of the possible fixed points.

One can perform an analysis (linearization) in the neighborhood of such a fixed point which shows how the renormalization transformation acts on points which differ only slightly from the fixed point itself. This yields local axes which determine how lengths are scaled in different directions around the fixed point. In turn, this analysis allows for the calculation of the critical exponents. The fixed point describes, therefore, a specific critical behavior whose salient properties are determined by the local behavior of the renormalization group

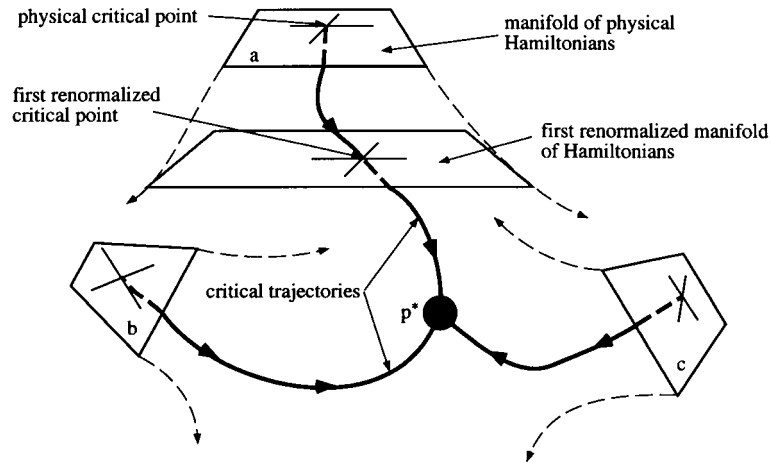


Fig 7. Universality of critical phenomena. Critical trajectories are bold lines. p^* is a common fixed point for the three distinct physical critical points on the manifolds a , b , and c .

transformation around the fixed point. *Furthermore*, and this is where we finally get an understanding of the universality of this critical behavior, *corresponding to each fixed point is a domain of attraction. These are all those points in the abstract parameter space (the space of all Hamiltonians) which, under the transformation τ , eventually flow into the fixed point.*

How does this provide an explanation for the universality of critical phenomena? Figure 7 helps indicate how.²⁶ Suppose that the manifold a of physical Hamiltonians represents different states of nickel near its ferromagnetic critical point.

Under the renormalization transformation the points on the manifold are mapped to other points in the parameter space. The physical critical point (corresponding to the nickel at the phase transition) has infinite correlation length and so, as we have seen, it gets mapped to another point with infinite correlation length. This defines the critical trajectory which is the same as the line from p to p^* in Figure 4. Thus it lies on the critical surface. Suppose that physical manifold b corresponds to some other metal, say, iron; and likewise that physical manifold c corresponds to some third material, gadolinium. Notice that the critical trajectories from the physical manifolds a , b , and c all flow to the same fixed point p^* . They must, therefore, all lie on the critical surface S_∞ . Now, recall that the critical behavior—in particular, the set of critical exponents—is a function of the nature of the renormalization group

²⁶ The 'physical line' of Figure 3 and 4 is a line on this physical manifold with an end-point at the physical critical point.

flow near the relevant fixed point. Thus, nickel, iron, and gadolinium must all have their phase transitions characterized by the same critical exponents: *They all lie in the basin of attraction of the same fixed point*. This is the key to understanding the universality of critical phenomena.

Acknowledgements

I would like to thank my colleagues at Ohio State for helpful discussions and comments. Special thanks to Carolyn McMullen, Justin D'Arms, Joe Mendola, Dion Scott-Kakures and Mark Wilson. Extra special thanks to Diana Raffman and William Taschek for all of their help and patience.

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